

RecRAG five-record construction and retrieval report

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2026-09-09

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Read this first

The goal was simple. Build a fact graph from text, search that graph, and see whether the system can answer questions.

The run used five questions from the 2WikiMultihopQA development data. Each question came with 10 source pages. Each record got its own graph and its own question. This avoids mixing facts from different questions.

The run compared two ways to build the graph and four ways to search it:

- **Graph construction** means turning text into named things and links between them.
- **Retrieval** means finding the pieces of text or graph data that may answer a question.
- **Answer generation** means writing the final answer from the retrieved data.
- **Scoring** means using fixed checks and an LLM judge to give each result a number from 0 to 1.

Main result

Ontology-guided construction made better graphs in this five-record run. Its average groundedness was 0.78 versus 0.56 for Standard. Its average completeness was 0.58 versus 0.44. Its average supporting-evidence coverage was 0.64 versus 0.34.

For question answering, Vector and Hybrid retrieval were the most steady. They reached 1.00 average contextual recall for both construction methods. Text2Cypher often returned no context, especially from the ontology-guided graphs.

This is a small run. These numbers describe these five records, not every possible question.

Run at a glance

Item	Value
Data	2wikimulti-hopqa/dev.json
Records	5
Questions per record	1
Source pages per record	10
Graph construction methods	Standard, ontology-guided
Retrieval methods	Text2Cypher, agentic, vector, hybrid
Graphs built	10
Retrieval and answer cases	40
Run errors	0
Chat and judge model	gpt-5.6-terra, reasoning effort xhigh
Embedding model	text-embedding-3-small, 1,536 dimensions
Retrieved items scored	Top 5

Small dictionary

- **Entity.** A named thing, such as a person, film, place, or organization.
- **Relationship.** A named link between two entities, such as DIRECTED or BORN_IN.
- **Graph triple.** One subject, one relationship, and one object. For example, Person – BORN_IN – Place.
- **Chunk.** A section of source text. The run used chunks of 1,000 characters with 100 characters of overlap.
- **Embedding.** A list of numbers that represents the meaning of text. The system uses it to find similar text.
- **Vector search.** Search based on embedding similarity.
- **Full-text search.** Search based on words and phrases in the text.
- **Cypher.** Neo4j's query language. Text2Cypher asks the model to write a read-only Cypher query.
- **Agentic retrieval.** The agent chooses a search tool and may run a second search after reading the first result.
- **LLM judge.** A language model that reads a result and scores it against written rules.

What the system tested

The five records

Record	Question	Expected answer
1	Who is the mother of the director of film Polish-Russian War (Film)?	Małgorzata Braunek
2	Which film came out first, Blind Shaft or The Mask Of Fu Manchu?	The Mask Of Fu Manchu
3	When did John V, Prince Of Anhalt-Zerbst's father die?	12 June 1516
4	What is the award that the director of film Wearing Velvet Slippers Under A Golden Umbrella won?	Myanmar Motion Picture Academy Awards
5	Where was the director of film Ronnie Rocket born?	Missoula, Montana

The two construction methods

Standard construction uses the default Neo4j GraphRAG extraction prompt. The model decides which entity types and relationship names to use.

Ontology-guided construction gives the model a short list of entity types and relationship types. The model can still add types when the text needs them.

Both methods write the graph, chunk embeddings, and full-text index to Neo4j.

The four retrieval methods

Text2Cypher. The model turns the question into a read-only graph query.

Vector. The system searches the chunk embeddings for text with a similar meaning.

Hybrid. The system combines embedding search and full-text search.

Agentic. An agent chooses between vector search and Text2Cypher. It can refine the search once.

The answer step

The system gave the retrieved context to the chat model. The model wrote an answer. The evaluator then compared that answer with the expected answer and the supporting passages.

How to read the scores

Every judge score runs from 0 to 1. Higher is better. A score of 0.80 means the judge found most of the expected behavior. It does not mean that 80 percent of all facts were correct unless the metric says so.

Construction scores

- **Groundedness.** Are the facts in the graph supported by the source pages? The judge also checks whether entities and relationship directions make sense.
- **Completeness.** How much factual information from all source pages made it into the graph? This check does not look only at the question.
- **Supporting evidence coverage.** How much information from the record's two supporting passages made it into the graph?

Graph structure measures

- **Entity count.** Number of entity nodes.
- **Relationship count.** Number of links between entity nodes. In this report it also equals the number of graph triples read by the evaluator.
- **Relation type count.** Number of different relationship labels.
- **Isolated entity count.** Number of entity nodes with no links.
- **Duplicate entity name groups.** Number of names used by more than one entity node.
- **Duplicate entity nodes.** Total nodes inside those duplicate-name groups.
- **Self-loop count.** Number of links where an entity points to itself.

Retrieval scores

- **Contextual precision.** Do useful results appear above less useful results?
- **Contextual recall.** Did the retrieved context contain the information needed from the supporting passages?
- **Contextual relevancy.** How much of the retrieved context relates to the question? A result can have high recall and low relevancy when it finds the needed facts among many distractors.

Answer scores

- **Answer relevancy.** Does the answer address the question?
- **Answer correctness.** Does the answer match the question, expected answer, or a listed alias? This is an LLM judge score and may be partial, such as 0.90.
- **Faithfulness.** Are the answer's claims supported by the retrieved context? No score is given when retrieval returns no context.
- **Alias match.** A strict local check. After simple text cleanup, the whole answer must equal the expected answer or one listed alias. A longer answer can be correct and still fail this check.

Construction results

Average scores

Method	Groundedness	Completeness	Supporting evidence coverage
Standard	0.56	0.44	0.34
Ontology-guided	0.78	0.58	0.64

Scores by record

Record	Standard G / C / S	Ontology-guided G / C / S
1	0.60 / 0.30 / 0.70	0.90 / 0.70 / 0.70
2	0.30 / 0.30 / 0.20	0.50 / 0.60 / 0.20
3	0.90 / 0.90 / 0.70	0.90 / 0.70 / 0.90
4	0.30 / 0.30 / 0.10	0.90 / 0.80 / 0.70
5	0.70 / 0.40 / 0.00	0.70 / 0.10 / 0.70

G / C / S means groundedness, completeness, and supporting evidence coverage, in that order.

Graph size and quality checks

Method	Avg build seconds	Avg triples	Avg entities	Avg relation types	Avg duplicate groups	Avg duplicate nodes	Avg self-loops
Standard	30.74	46.40	55.20	8.80	0.00	17.60	0.00
Ontology-guided	41.45	70.80	65.80	26.40	0.60	1.20	0.60

Method	Record	Seconds	Triples	Entities	Relationships	Relation types	Isolated	Duplicate groups	Duplicate nodes	Self-loops
Standard	1	40.69	40	62	40	11	0	0	17	0
Standard	2	32.67	55	60	55	10	0	0	25	0
Standard	3	27.06	64	34	64	6	0	0	0	0
Standard	4	26.51	38	52	38	10	0	0	19	0
Standard	5	26.76	35	68	35	7	0	0	27	0
Ontology-guided	1	24.75	63	69	63	34	0	1	2	1
Ontology-guided	2	51.42	89	68	89	36	0	2	4	1
Ontology-guided	3	36.59	59	38	59	10	0	0	0	0
Ontology-guided	4	57.67	65	68	65	28	0	0	0	1
Ontology-guided	5	36.82	78	86	78	24	0	0	0	0

Retrieval and answer results

Average scores

Construction	Retrieval	C. precision	C. recall	C. relevancy	Answer relevancy	Correctness	Faithfulness	Alias match
Standard	text2cypher	0.60	0.30	0.60	1.00	0.60	0.67	1/5
Standard	agentic	0.87	1.00	0.13	1.00	1.00	1.00	0/5
Standard	vector	0.94	1.00	0.14	1.00	1.00	1.00	1/5
Standard	hybrid	0.92	1.00	0.13	1.00	1.00	1.00	1/5
Ontology-guided	text2cypher	0.20	0.00	0.20	0.20	0.20	1.00	0/5
Ontology-guided	agentic	0.91	0.80	0.21	1.00	0.98	0.80	0/5
Ontology-guided	vector	0.86	1.00	0.19	1.00	0.98	1.00	0/5
Ontology-guided	hybrid	0.94	1.00	0.12	1.00	0.98	1.00	1/5

Faithfulness uses only cases with retrieved context. A dash means that no case in that row had a faithfulness score.

Per-record scores

Construction	Retrieval	Record	C. precision	C. recall	C. relevancy	Answer relevancy	Correctness	Faithfulness	Alias
Standard	text2cypher	1	1.00	0.00	1.00	1.00	1.00	1.00	yes
Standard	text2cypher	2	1.00	0.50	1.00	1.00	1.00	1.00	no
Standard	text2cypher	3	1.00	1.00	1.00	1.00	1.00	0.00	no
Standard	text2cypher	4	0.00	0.00	0.00	1.00	0.00	-	no
Standard	text2cypher	5	0.00	0.00	0.00	1.00	0.00	-	no
Standard	agentic	1	0.81	1.00	0.19	1.00	1.00	1.00	no
Standard	agentic	2	0.76	1.00	0.08	1.00	1.00	1.00	no
Standard	agentic	3	1.00	1.00	0.17	1.00	1.00	1.00	no
Standard	agentic	4	0.87	1.00	0.11	1.00	1.00	1.00	no
Standard	agentic	5	0.92	1.00	0.14	1.00	1.00	1.00	no
Standard	vector	1	1.00	1.00	0.20	1.00	1.00	1.00	yes
Standard	vector	2	0.83	1.00	0.07	1.00	1.00	1.00	no
Standard	vector	3	1.00	1.00	0.17	1.00	1.00	1.00	no
Standard	vector	4	1.00	1.00	0.10	1.00	1.00	1.00	no
Standard	vector	5	0.87	1.00	0.17	1.00	1.00	1.00	no
Standard	hybrid	1	1.00	1.00	0.18	1.00	1.00	1.00	yes
Standard	hybrid	2	0.76	1.00	0.09	1.00	1.00	1.00	no
Standard	hybrid	3	1.00	1.00	0.14	1.00	1.00	1.00	no
Standard	hybrid	4	1.00	1.00	0.10	1.00	1.00	1.00	no
Standard	hybrid	5	0.87	1.00	0.14	1.00	1.00	1.00	no
Ontology-guided	text2cypher	1	0.00	0.00	0.00	0.00	0.00	-	no
Ontology-guided	text2cypher	2	1.00	0.00	1.00	1.00	1.00	1.00	no
Ontology-guided	text2cypher	3	0.00	0.00	0.00	0.00	0.00	-	no
Ontology-guided	text2cypher	4	0.00	0.00	0.00	0.00	0.00	-	no
Ontology-guided	text2cypher	5	0.00	0.00	0.00	0.00	0.00	-	no
Ontology-guided	agentic	1	0.81	1.00	0.16	1.00	1.00	1.00	no
Ontology-guided	agentic	2	1.00	0.00	0.50	1.00	1.00	0.00	no

Construction	Retrieval	Record	C. precision	C. recall	C. relevancy	Answer relevancy	Correctness	Faithfulness	Alias
Ontology-agentic guided		3	0.83	1.00	0.16	1.00	1.00	1.00	no
Ontology-agentic guided		4	1.00	1.00	0.11	1.00	0.90	1.00	no
Ontology-agentic guided		5	0.92	1.00	0.11	1.00	1.00	1.00	no
Ontology-vector guided		1	0.64	1.00	0.22	1.00	1.00	1.00	no
Ontology-vector guided		2	0.95	1.00	0.24	1.00	1.00	1.00	no
Ontology-vector guided		3	0.83	1.00	0.18	1.00	1.00	1.00	no
Ontology-vector guided		4	1.00	1.00	0.17	1.00	0.90	1.00	no
Ontology-vector guided		5	0.89	1.00	0.14	1.00	1.00	1.00	no
Ontology-hybrid guided		1	1.00	1.00	0.14	1.00	1.00	1.00	yes
Ontology-hybrid guided		2	1.00	1.00	0.04	1.00	1.00	1.00	no
Ontology-hybrid guided		3	0.83	1.00	0.19	1.00	1.00	1.00	no
Ontology-hybrid guided		4	1.00	1.00	0.12	1.00	0.90	1.00	no
Ontology-hybrid guided		5	0.87	1.00	0.10	1.00	1.00	1.00	no

What the numbers say

- The ontology-guided prompt produced better graph scores on all three construction checks. The largest gap was supporting evidence coverage, 0.64 versus 0.34.
- Standard Vector and Hybrid retrieval both had 1.00 contextual recall and 1.00 answer correctness on average.
- Standard Vector and ontology-guided Hybrid tied for the highest average contextual precision at 0.94.
- Text2Cypher was the weakest retrieval method in this run. It often returned no context, which led to low recall and low answer correctness.
- Vector and Hybrid found the needed information, but their contextual relevancy scores stayed between 0.12 and 0.19. They also returned unrelated text.
- Agentic retrieval gave strong answers. Its ontology-guided contextual recall was 0.80 because one record missed part of the needed context.
- The alias check is strict. It marked many longer, correct answers as a non-match.

Limits and run notes

This report covers five records. The LLM judge can produce a different score on another run. The results should be read as a comparison inside this run, not as a final ranking for all data.

The final run used 10 fresh Neo4j databases, one for each record and construction method. Each database was online and had the required graph indexes. Neo4j printed deprecation warnings for existing subquery syntax, but no case failed.

The first database setup check was discarded before scoring. It is not included in any table in this report.